Reference Event 1: The Bombing of the Alfred P. Murrah Federal Building, Oklahoma City, 19 April 1995

Characterization Against the Government Facilities Sector Design Basis

DBA-GF-RE-1 — Reference Event Instrument, DBA-GF Series

Lineage: Issued under the Government Facilities Sector Framework v0.4; DBA-MA Parent Framework; conferred on the proposal of Government Operations and Continuity (DBA-GF-GO-FC1)

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Timothy E. Roxey

Eclectic Technologies

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This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

1. Purpose, Scope and Position

1.1 Purpose

This instrument reads the bombing of the Alfred P. Murrah Federal Building in Oklahoma City on 19 April 1995 back through DBA-GF-GO-FC1 (*Government Operations and Continuity*), whose category of vehicle-borne explosive attack on an occupied building it characterizes (§5.2), at the class's first reference facility, the multi-tenant federal office building (§1.3). It asks whether the class's design basis anticipated what happened, and returns what it finds.

1.2 Position and Scope

A reference event instrument, in the series convention, reads a real event backward through the design basis, adds no requirements of its own, and returns findings and recommendations for disposition. This one is issued under the Government Facilities Sector Framework v0.4, inherits the DBA-MA parent framework, and was conferred by the framework's §9.1 on the author's decision of 14 October 2026, on the proposal of *Government Operations and Continuity*. Section references of the bare form §N are to that class at v0.1, the version this instrument read.

The scope is the attack, the building's performance, the harm to the people in and near it, and the government's response in the protection of federal buildings. The criminal record is closed. Those responsible were tried and convicted, and no one who committed or aided the attack is named. None of the dead is named. Officials and agencies are named by role. Nineteen of the dead were children; the instrument states what the record establishes about them and no more.

The building, its occupants and the governance functions it housed lie within this class. The Foundational Definitions are cited by designation, FD-REG holding their versions: FD-DBL (*Design Basis Layers*) bears on Section 5.

2. Sources and Attribution

2.1 Sources and Their Limit

The primary sources are the National Institute of Standards and Technology's summary of the investigation of the building's performance, which reports the work of the team the Federal Emergency Management Agency assembled; the epidemiological study of the casualties by the Oklahoma State Department of Health, *Physical Injuries and Fatalities Resulting From the Oklahoma City Bombing*, published in the *Journal of the American Medical Association* in August 1996; the Department of Justice's *Vulnerability Assessment of Federal Facilities* of 28 June 1995; Executive Order 12977 of 19 October 1995; and the Congressional Research Service's report RS22121 on the Interagency Security Committee. The continuity of the tenant agencies' functions after the attack is not set out in any source read.

2.2 Where the Sources Differ

The class gives the injured as more than five hundred on the Associated Press's count. The Department of Health's study counts 759 injured, of whom 83 were hospitalized and 509 treated as outpatients; this instrument rests on the study. The study counts 167 deaths among those present; the figure of 168 used by the class and by most accounts is the official count, and the instrument does not reconcile the two, its readings depending on neither. The investigating team's estimate of the charge, about 4,000 pounds of TNT equivalent, is an estimate and is stated as one.

3. The Event

3.1 The Attack

At about nine in the morning on 19 April 1995 a truck bomb was detonated at the curb on the north side of the Murrah building. On the investigating team's estimate it was equivalent to about 4,000 pounds of TNT and stood about sixteen feet from the nearest column. The building housed field offices of several federal agencies, among them the Social Security Administration, the Department of Housing and Urban Development, the Department of Veterans Affairs, the Drug Enforcement Administration and the Bureau of Alcohol, Tobacco and Firearms, with a federal employees' credit union, military recruiting offices and a day-care center on the second floor.

3.2 The Building's Performance

On NIST's summary, the blast destroyed the columns that supported a transfer girder at the third floor, and the loss of that support brought down the floors above in progressive collapse; roughly half of the building's usable space collapsed. The building was an ordinary reinforced-concrete moment frame designed to the codes of its time, which the team found had been applied adequately and which did not ask the building to resist blast. The team found that an ordinary frame could not redistribute its loads once the columns were lost, and estimated that the detailing used for special moment frames in seismic regions would have reduced the area that collapsed by between half and four-fifths.

3.3 The People

On the Department of Health's study, 361 people were in the federal building. Of them 319 were injured and 163 died, nineteen of them children. Of the 175 people in the part of the building that collapsed, 153 died; of the 186 elsewhere in the building, 10 died. Three people died in adjacent buildings and one outdoors, and injury rates in the adjacent buildings ran from 38 to 100 percent.

3.4 The Response

The Department of Justice's assessment of 28 June 1995 sorted federal facilities into five security levels and set minimum standards for each. Executive Order 12977 of 19 October 1995

established the Interagency Security Committee to set security standards for federal facilities, which it still does.

4. Technical Decomposition

The event decomposes into initiating elements (I-1 to I-3) and effect elements (E-1 to E-4), as Table 4-1 sets out. Composed, they are the class's vehicle-borne explosive attack (§5.2) at the multi-tenant federal office building, opening Chains L and G.

Table 4-1. Event elements and their mapping. Section references are to *Government Operations and Continuity* v0.1.

No.	What happened	Class provision	Section
I-1	Truck bomb at the curb of a public street, about sixteen feet from a column	Standoff (§3.3, §4.3)	3.1
I-2	Detonation during working hours, the building fully occupied	Occupant load (§3.3)	3.1
I-3	Columns under a third-floor transfer girder destroyed	Structural robustness (§3.3, §4.3)	3.2
E-1	Progressive collapse of about half the usable space	Structural robustness; Chain L	3.2
E-2	153 of 175 died in the collapsed part, 10 of 186 elsewhere	Chain L; first residual (§6.8)	3.3
E-3	Nineteen children among the dead; day-care center in the building	Those who cannot escape unaided (§3.3)	3.1, 3.3
E-4	Five security levels and the Interagency Security Committee	Regulatory context (§8.4)	3.4

5. Tier Determination

5.1 The Four Characteristics

Intentionality is present: the building was chosen and the attack planned. Persistence is not present: the attack was a single act, and no campaign followed it. Precision is present in part: the time and place were chosen, though nothing in the record shows the attack was aimed at the structure's critical elements. Denial is not present: nothing in the record shows a design to keep the government from governing for a time that mattered to any conflict.

5.2 Determination

The event does not meet the threshold for the military-action extension. Under the author's rulings of 10 and 13 October 2026, recorded at FD-DBL (*Design Basis Layers*) §6.4, all four characteristics must be established, and persistence and denial are not. The attack was a domestic act of terrorism, and no state was involved. It is recorded as a non-military adversarial event

realizing the class's §5.2 in Chains L and G. That is the tier determination returned for the DBA Reference Event Register. No impacted sector beyond Government Facilities is proposed.

6. Characterization Against the Design Basis

6.1 §2.4 and §5.2 The Class's Account of the Event

Held, with a gap. The account holds against NIST's summary and the federal record: the time, the charge as estimated, the transfer girder and the progressive collapse, the finding on the codes, the estimate for seismic detailing, the tenant agencies, the assessment of June 1995 and the executive order. Erratum E-1: the injured are given as more than five hundred on the Associated Press's count; the Department of Health's study counts 759, and the class should give that figure with its source (Section 2.2). Two refinements are offered and are not errata: the class may add the study's division of the dead between the collapsed part and the rest of the building, which is the strongest evidence for its own reading; and the bomb's distance of about sixteen feet from the nearest column, which states the standoff the class discusses.

6.2 §4.3 Standoff and the Structure

Held. The class holds that standoff and structural robustness are both design-basis quantities and that neither substitutes for the other. The record bears it out with unusual clarity. The standoff was the width of a curb lane. Of those in the part of the building that collapsed, 87 percent died; of those in the part that stood, 5 percent died. The blast set where the structure failed; the structure's failure set who died. Finding F-1.

6.3 §3.3 The Occupant Load and Those Who Cannot Escape Unaided

Held, with a gap. The class states the occupant load by kind, including those who cannot escape unaided. The record shows that what mattered was not how many such occupants a building holds but where they are held in relation to the face a vehicle can reach and the part of the structure that depends on the elements a blast there can remove. The class's parameter counts them and does not place them. Finding F-2.

6.4 §5.2 The Event Category

Held. The category's postulated event, consequence threshold and required function match the event. The category was written from it.

6.5 §4.2 The Functions Housed

Not exercised. On the record read, the tenant agencies lost offices, records and people, and their functions were performed again; how, from where and in what time, no source read sets out, so the relocation time and the alternate site are not read here. Open item OI-3.

6.6 §8.4 The Protection of Federal Buildings

Held. The class traces the facility security levels and the Interagency Security Committee to this event, and the Department of Justice's assessment and Executive Order 12977 confirm it.

6.7 §6.8 The First Declared Residual

Held. An attack that kills people before any protection acts leaves a residual in lives that no restoration reduces. The event is that residual, and its measure, on the study's figures, was set by the collapse and not by the blast alone.

6.8 §7.3 The AI-Governance Design Basis

Not exercised. No component within the scope of *Layer Decomposition and Determinism* §4.3 bore on the event.

6.9 Summary of Design Basis Performance

Of the eight readings, two Held, with a gap, the class's account among them for its one erratum; four Held; and two were Not exercised. Nothing was Missed. The class was written from this event and reads it. What the event adds is a measure: structural robustness can be stated as the people within the region that depends on any element a reachable blast can remove, and those who cannot escape unaided must be placed, not only counted.

7. Findings

Section references are to *Government Operations and Continuity* v0.1 unless another instrument is named.

7.1 F-1 — Structural Robustness Is Measured by the People It Protects

Of 175 people in the part of the building that collapsed, 153 died; of 186 in the part that stood, 10 died (Section 3.3). The finding confirms §4.3, that standoff and structure are each required, and is that the class should state the consequence of the structural-robustness parameter in occupant terms: for each structural element a vehicle-borne charge at the facility's stated standoff could remove, the floor area whose support depends on it and the occupant load within that area at the facility's most occupied hour. A first- or second-tier facility whose standoff falls short of its assigned threat would state that figure and what it does to reduce it.

Affected: §3.3, §4.3, §5.2, §6.7. Disposition: R-1.

7.2 F-2 — Those Who Cannot Escape Unaided Are Placed, Not Only Counted

Nineteen of the dead were children, in a building that housed a day-care center (Sections 3.1 and 3.3). The finding is that the class should require a first- or second-tier facility to state where in the building those who cannot escape unaided are held, among them children in day care, in

relation to the faces a vehicle can reach and to the floor area of F-1, and to treat their placement there as a shortfall to be stated with what is done about it, as §4.3 already requires for the relocation of the most exposed occupants.

Affected: §2.1, §3.3, §4.3. Disposition: R-2; noted to the coordination class, DBA-GF-CntlCo, for the tenancy standards.

8. Recommendation Register

Priority is on the series' one-to-three scale, 1 the most consequential. R-1 and R-2 dispose to *Government Operations and Continuity*.

No.	Title	Affected	Recommendation	Pri.
R-1	State robustness in occupants	§3.3, §4.3, §5.2, §6.7	For each element a blast at the stated standoff could remove, state the dependent floor area and its occupant load at the most occupied hour.	1
R-2	Place the most vulnerable	§2.1, §3.3, §4.3	State where those who cannot escape unaided are held relative to reachable faces and the area of R-1; treat exposure there as a stated shortfall. Noted to CntlCo.	1

9. Open Items

OI-1: the register row. Project 0 is asked to enter the tier determination of Section 5.2. OI-2: erratum E-1 is owed to the class at its next revision. OI-3: the continuity of the tenant agencies' functions after the attack is not read; if a source sets it out, Section 6.5 is revised.

10. Standing of This Document

10.1 What This Instrument Establishes

It modifies neither class nor framework, and makes no finding of fault. It rests on the government's findings, stated as their own.

10.2 Relationship to the Class's Use Case

The class's use case traces §6.3, the seat struck during the conflict. This event bears on its life-safety half: a single charge at a public curb, an ordinary frame that could not redistribute its loads, and a death rate about sixteen times higher in the part that fell than in the part that stood. F-1 gives the use case the measure on which to state what a strike on an occupied seat costs in people. How a strike would be delivered is the use case's construction.

10.3 Relationship to Other Instruments

The Brussels Airport and Maalbeek Bombings (DBA-TRAN-RE-5) reads explosive attacks on occupied public spaces. *The Breach of the United States Capitol* (DBA-GF-RE-2) and *The Ransomware Attack on the City of Atlanta* (DBA-GF-RE-3) read the class's other two reference facilities.

References

National Institute of Standards and Technology, *Oklahoma City Bombing (1995)*, summary of the investigation of the building's performance. Mallonee and others, Oklahoma State Department of Health, *Physical Injuries and Fatalities Resulting From the Oklahoma City Bombing*, Journal of the American Medical Association 276 (1996). United States Department of Justice, *Vulnerability Assessment of Federal Facilities*, 28 June 1995. Executive Order 12977, *Interagency Security Committee*, 19 October 1995. Congressional Research Service, *The Interagency Security Committee and Security Standards for Federal Buildings* (RS22121). The DBA instruments named in the text, with the Foundational Definitions and FD-REG, and the DBA-MA Parent Framework.

Revision History

Version 0.1, 14 October 2026: issued in Project 11 under the Government Facilities Sector Framework v0.4, on the conferral of the framework's §9.1 of 14 October 2026, read against *Government Operations and Continuity* v0.1. Written new under the Reference Event Compression Standard v0.3. Tier determination at Section 5.2, under FD-DBL §6.4 as ruled on 10, 13 and 14 October 2026; F-1 and F-2, R-1 and R-2; erratum E-1 to the class.